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Active learning for noisy physical experiments with more than two responses

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ABSTRACT

When designing experiments, one of the objectives consists in building a model for prediction of future observations. If the underlying phenomenon is complex, one option is to employ machine learning (ML) models for data analysis and to build accurate emulators that function as virtual representations of the physical process, and can be used in lieu of further evaluations of the actual physical system. However, to obtain accurate models, informative data must be provided to train the algorithms, and a typical approach consists in the sequential collection of data via active learning (AL) strategies. Most existing literature on AL focuses on computer experiments. In this paper we introduce an AL algorithm for Physical Experiments based on nonparametric Ranking and Clustering (ALPERC) that can be used for sequential data collection in noisy settings when three or more responses are investigated in the same experiment. We inspect the performance of the ALPERC algorithm through simulations and a case study application on the prediction of relevant properties of thermoelectric materials.

1. Introduction

In the machine learning (ML) context, active learning (AL) is a type of supervised learning technique devoted to the iterative selection of the most informative samples with the aim of maximizing information acquisition [1]. Consequently, the AL objective is to maximize a criterion, typically in terms of *optimization* or *prediction*, with a limited number of data points. This closely relates to Design of Experiments (DOE) that traditionally builds an empirical model of a process given few observations, with the ultimate objective of driving business decisions [2]. The iterative nature of AL interweaves with the concepts of "adaptive sampling" and "sequential design" from the DOE literature. In all cases the idea is that, given the information provided by current samples, by adaptively (i.e. sequentially) exploring the most informative experimental configurations, the objective of the project can be fulfilled employing less time and resources.

AL has recently been employed in many different industries, proving its effectiveness in disparate contexts [3–13]. One categorization of such applications can be carried out in relation to the sampling strategies adopted by the AL procedure. Those strategies depend on the final objective of the study: most applications focus on the detection of some specific features, typically the global optimum, leading to the selection of design points that are grouped around specific regions of the design space. We refer to this objective as *optimization*. On the other hand, the goal might be to generate a model capable of accurately emulating the specific phenomenon within the whole design space. In this case the design points should be, at least to a certain extent, spread out in order to learn the behavior of the function in the whole region of interest. This is the *prediction* objective.

The general framework of the AL methodology consists on a few fundamental building blocks, that are:

1. An initial dataset $\mathbf{\Omega}_0 = [\mathbf{D}_0\ \mathbf{Y}_0]$ where $\mathbf{D}_0$ is the matrix that includes n_0 samples that are combinations of the d input features $\mathbf{D}_0 = \{\mathbf{x}_1, \ldots, \mathbf{x}_{n_0}\}$, with $\mathbf{x} = (x_1, \ldots, x_d)$ one of the input vectors, and the matrix of responses $\mathbf{Y}_0 = \{\mathbf{y}_1, \ldots, \mathbf{y}_{n_0}\}$ with $\mathbf{y} = (y_1, \ldots, y_c)$ one vector of the c responses corresponding to the input vector $\mathbf{x}$. The response can be in the form of either matrix ($c > 1$) or vector ($c = 1$).
2. c models $f_1(\cdot), \ldots, f_c(\cdot)$ that are trained on $\mathbf{\Omega}_0$ and can provide estimation of a generic realization $\hat{\mathbf{y}}$.
3. A criterion that uses the models $f(\cdot)$ to suggest the next experimental configurations $\mathbf{D}_{add,0}$ at which the response(s) must be evaluated. In the *optimization* literature, this is often called "acquisition function" or "infill criterion" [14].

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When $\mathbf{D}_{add,0}$ is found and the corresponding $\mathbf{Y}_{add,0}$ are acquired, the matrix $\mathbf{\Omega}_{add,0} = [\mathbf{D}_{add,0}\ \mathbf{Y}_{add,0}]$ is concatenated to $\mathbf{\Omega}_0$: $\mathbf{\Omega}_1 = \begin{bmatrix} \mathbf{\Omega}_0 \\ \mathbf{\Omega}_{add,0} \end{bmatrix}$ and the steps above are iterated until a stopping condition is met.

The underlying phenomenon under investigation can be either an actual physical process or a complex computer emulator. In both cases the goal is to accurately predict or optimize the response(s) of interest by using the fewest possible resources. However, some relevant differences can also be pointed out. For instance, physical experiments are affected by noise, while computer emulators are often deterministic in nature. Another difference regards the choice of the initial design $\mathbf{D}_0$: in the case of computer experiments, space-filling designs can be selected to favor a precise estimation of the response, while for physical experimentation this may not be feasible as a large number of factor levels can be too costly or unfeasible in practical applications.

In this paper we propose a new AL algorithm that can drive sequential experimentation when the objective is the accurate prediction of some responses of interest. We focus on the case of physical experimentation when multiple responses ($c \geq 3$) must be accurately predicted by employing the same set of data.

The remainder of the paper is organized as follows. Section 2 briefly presents previous contributions related to this study; section 3 describes the desired properties of AL for physical experiments; section 4 provides information on the essential building blocks and explains the methodology developed; and section 5 focuses on an illustrative example in which we describe our method. We then demonstrate the performance of the proposed methodology against competitors in a simulation study (section 6) and we present a case study involving thermoelectric materials (section 7). Finally, in section 8 a summary of the main contributions of the work and some concluding remarks are presented.

2. Literature background

In this section we briefly discuss the main contributions in relation to algorithms for the selection of experimental configurations to run in an AL framework. We focus on the studies in which the target is global accuracy of the predictive models, but extensive literature also exists on the optimization of black-box functions from both computer and physical experiments. For a recent review of these topics, please refer to Ref. [14].

Active learning MacKay (ALM) and active learning Cohn (ALC) are two of the earliest algorithms proposed in the literature for the emulation of complex functions by sequential data collection, inspiring several subsequent authors. ALM favors the selection of those points that have the highest uncertainty, and aims to maximize the expected information gain based on Shannon's entropy [15]. Thanks to its intuitiveness and ease of implementation, ALM has been widely adopted, also with slight modifications, as a method for data acquisition in AL (e.g. Refs. [11,16]). ALC on the other hand chooses those points that minimize the expected integrated variance over the entire input space [17]. This translates into the selection of those points $\mathbf{x}'$ that maximize the expected reduction in predictive error over the input space [18]:

$$\int [\sigma^2(\mathbf{x}) - \sigma^2_{\mathbf{x}'}(\mathbf{x})]\, d\mathbf{x} \tag{1}$$

where $\sigma^2(\mathbf{x})$ is the variance observed over the input space given the points in $\mathbf{D}$, and $\sigma^2_{\mathbf{x}'}(\mathbf{x})$ is the expected predictive variance had the new configuration $\mathbf{x}'$ been added to $\mathbf{D}$. In practice the integral is often replaced by a sum over a grid of candidate locations [18].

A more recent contribution uses ALM and ALC in combination with Bayesian partition models to guide the selection of the subsequent experimental configurations [18]. The procedure obtains separate sequential maximum entropy designs in each area of the design space, according to the partitions identified by a treed Gaussian Process (GP) model, as such enabling the development of highly flexible nonstationary models that can learn complex behaviors even in the presence of heteroscedastic noise structures [18]. Another approach, built specifically to deal with input-dependent noise structures, is presented by Ref. [19]. In their work, they propose a greedy algorithm that sequentially adds one input configuration at a time balancing the exploration of new input combinations with the exploitation of current knowledge by the inclusion of replicates of existing combinations, also considering a lookahead perspective that aims to minimize the integrated mean squared prediction error over a horizon of sequential design iterations. The replication is used to effectively separate signal from noise, and it has the additional advantage of offering computational savings for the heteroscedastic GP model implementation [20] used in Ref. [19]. [21] propose an AL methodology based on ALC and Laplacian regularization that is suited to problems involving the estimation of noisy black-box functions. The procedure uses the GP model and selects the next most informative configuration from a large set of candidate points based on a space-filling design. The main strengths of the procedure include the ability to use the information from both evaluated and unevaluated points for a principled selection of the next experimental configuration to add. Interestingly, few contributions focus on the impact of the initial design choice on the goodness of the AL procedure. One relevant exception is the work of [22]; that shows how common space-filling designs such as Latin hypercube sampling (LHS) and maximin may perform worse than a random design when the predictive model is a GP. Consequently, the authors develop the betadist designs where the distances among experimental configurations in the design space are Beta-distributed, thus favoring an effective estimation of the GP parameters. Moreover, they also develop a hybrid LHS-betadist design, and show how this can outperform other competitors from the literature when AL based on ALM is employed.

In general, each contribution briefly presented in this section has its own peculiarities and strengths, but they all share one characteristic: they are developed to be used for the analysis of computer experiments. Instead, the purpose of this paper is to develop an AL framework that can be applied to physical experiments, thus specific properties and requirements are needed, and, to the best of our knowledge, none of the examined methods directly satisfies all of them. The next section aims to specify the desired properties of the new AL framework.

3. Desired properties of AL for physical experiments

Computer experiments and physical experiments have different peculiarities attributable to their diverse settings. Here, therefore, we specify the desired properties of the AL framework when the objective is to accurately predict the underlying functions within the whole design space, that is to build an emulator that can be used in lieu of further evaluations of the physical phenomenon by means of execution of experiments. The desired properties are:

1. The use of an adequate experimental design: in the literature on computer experiments, space-filling designs are used [23]. Unfortunately, they have one characteristic that hinders their applicability to physical experiments, that is they have as many factor levels as runs in the design. This is not feasible in practical applications of physical experimentation as it can be too costly, if not impossible, to evaluate a large number of factor levels. Another solution is needed.
2. The selection of the most informative configurations to run in batch. While for computer experiments it may be convenient to run trials one at a time, in physical experimentation it is usually more practical to run experiments in batches. This makes it possible to conduct tests in parallel, and may be required in those companies that adopt strict procedures and protocols for experimentation, or where experimentation interferes with production. Sequential experimentation has been reported to be one of the open issues in DOE [24], and we believe that practitioners may be more willing to adopt a sequential strategy if new runs can be executed in batches.

3. The ability to treat multiple responses at the same time. It is not unusual to conduct experiments with the objective of studying the effects of the factors on multiple responses (e.g. Refs. [25–27]). Therefore, we consider this aspect and develop a strategy to be used when several responses are studied in the same experiment.
4. The ability to deal with noisy experiments. Although stochastic computer solvers have been proliferating in recent years (see section 2), computer emulators are, in large part, deterministic. On the contrary, physical phenomena are not deterministic and a certain level of noise is always present. This noise can be homoscedastic or heteroscedastic, and an AL strategy for physical experiments should be able to deal with both situations. An obvious implication is that the model chosen for the emulation of the physical process must be capable of providing an appropriate quantification of prediction uncertainty.

4. Methodology

In this section we first briefly review the predictive model and variable importance method used in this paper (section 4.1), and then we describe the AL methodology that we developed for application in physical experimentation (section 4.2). The proposed method is based on the combination of several different elements: (i) a MaxPro space-filling design with the factors restricted to the discrete numeric type; (ii) a predictive model to fit the observations and obtain a quantification of uncertainty of candidate points; (iii) a method to estimate the importance of variables in the models; (iv) a nonparametric ranking procedure to obtain a robust rank of the candidate points with respect to the uncertainty associated to each prediction; (v) a clustering algorithm that groups candidate points based on proximity in the design space. A scheme with a synthetic representation of the proposed methodology is provided in Fig. 1.

4.1. Predictive model and variable importance

In this paper we focus on the GP model as it is the most used in the AL literature and it can directly provide a quantification of uncertainty. This is required by the AL concept, that drives sequential acquisition by evaluating a measure of variance (see section 2). In the case of a deterministic setting, the GP assumption states that for a generic group of input locations $\mathbf{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, the corresponding outputs $\mathbf{Y} = (y_1, \ldots, y_n)^T$ follow a multivariate normal distribution:

$$\mathbf{Y} \sim \mathcal{N}(0, \tau^2\mathbf{K}) \tag{2}$$

where τ^2 is a multiplicative constant known as the scale parameter and $\mathbf{K}$ is the covariance matrix, whose entries are realizations of a kernel function $k(\cdot, \cdot)$ that depends upon the distance between the input vectors, and depends on a lengthscale parameter $\boldsymbol{\theta}$:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left\{ -\sum_{i=1}^{d} \frac{(x_i - x_i')^2}{\theta_i} \right\} \tag{3}$$

The one reported in (3) is the anisotropic Gaussian kernel, but several other options are available [28–30].

From theory on multivariate normal distributions it follows that given the training data $\boldsymbol{\Omega} = [\mathbf{D}\ \mathbf{Y}]$, the scale-adjusted predictions at n' unknown locations $\mathbf{D}' = \{\mathbf{x}_{n+1}, \ldots, \mathbf{x}_{n+n'}\}$: $Y(\mathbf{D}')|\boldsymbol{\Omega} \sim \mathcal{N}(\mu(\mathbf{D}'), \Sigma(\mathbf{D}'))$ have [30]:

$$\begin{array}{ll} \text{mean} & \mu(\mathbf{D}') = k(\mathbf{D}', \mathbf{D})\mathbf{K}^{-1}\mathbf{Y} \\ \text{and variance} & \Sigma(\mathbf{D}') = \tau^2[k(\mathbf{D}', \mathbf{D}') - k(\mathbf{D}', \mathbf{D})\mathbf{K}^{-1}k(\mathbf{D}', \mathbf{D})^T] \end{array} \tag{4}$$

The equations above depict one of the main advantages of GP models, namely a reliable quantification of uncertainty of predictions. The GP algorithm can be adapted to the case of input independent noisy data by adding a "nugget" term g to the diagonal elements of $\mathbf{K}$ [31]. In the case of heteroscedastic noise settings, the solution is not straightforward and involves, to describe the variance, the introduction of latent variables that are assumed to be multivariate normal under a GP prior. For further details, refer to Ref. [20] and their implementation of the heteroscedastic GP [32].

One of the characteristics that hinders the diffusion of complex ML models in industry is the difficulty to provide an explanation of the rationale behind the algorithms [33]. This is also the reason why ML models are often referred to as black-box models: the failure to articulate the rationale for an answer can affect the level of trust users will grant to the system [34]. An appropriate quantification of variable importance is the first step to explain the model's rationale, and can help in making the algorithm more interpretable for the analyst [34]. This also provides support for causal claims driven by the results of the predictive models. For these reasons, we introduce the concept of sensitivity analyses based on the Sobol' indices [35] for a quantification of variable importance in GP models. The Sobol' indices assume independent inputs and measure the relative importance of one input variable by the partial variance of model output explained by this variable [36]. Given an objective function $f(\mathbf{x})$ (e.g. a predictive model) Sobol's framework considers the functional decomposition:

$$\begin{aligned} f(x_1, \ldots, x_d) = f_0 + \sum_{i=1}^{d} f_i(x_i) + \sum_{i=1}^{d} \sum_{j>i}^{d} f_{ij}(x_i, x_j) + \cdots \\ + f_{1,\ldots,d}(x_1, \ldots, x_d) \end{aligned} \tag{5}$$

[35] shows that if the input variables are independent and $f(\cdot)$ is square integrable, from (5) the variance associated to the model outcome Y can be expressed as:

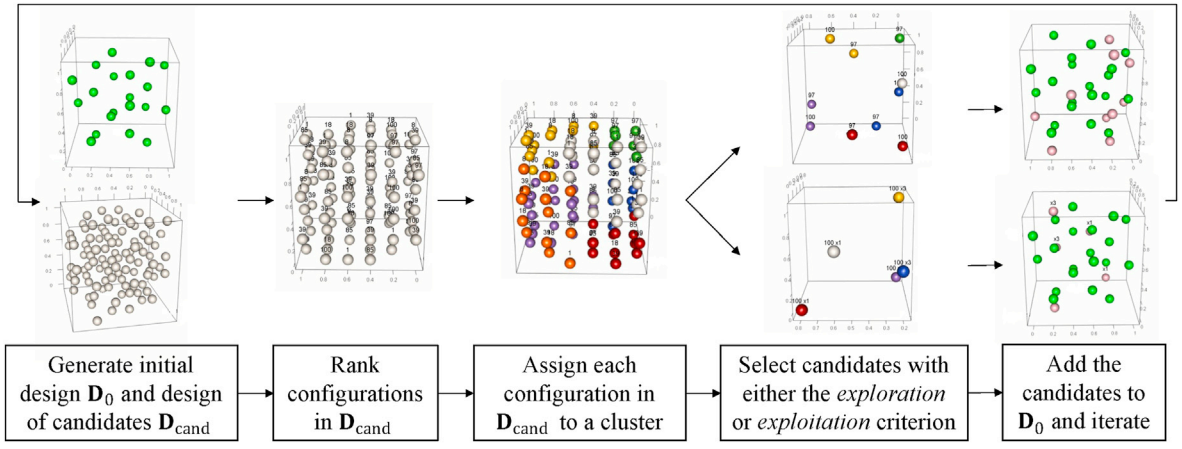


Fig. 1. Schematic representation of the proposed AL methodology.

$$V(Y) = \sum_{i=1}^{d} V_i + \sum_{i=1}^{d} \sum_{j>i}^{d} V_{ij} + \cdots + V_{1,\ldots,d} \tag{6}$$

where $V_i = V(\mathbb{E}(Y|x_i))$, $V_{ij} = V(\mathbb{E}(Y|x_i, x_j)) - V_i - V_j$ and so on. From this, the total partial variance V_{T_i} associated to the i-th input variable can be computed that considers not only the main effect of the i-th input, but also its interaction effects with all the other $d-1$ input variables $\mathbf{x}_{\sim i} = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_d)$ [36]. From (6), we obtain $V_{T_i} = V(Y) - V(\mathbb{E}(Y|\mathbf{x}_{\sim i}))$, thus the total sensitivity indices can be calculated as:

$$S_{Ti} = \frac{V_{T_i}}{V(Y)} = \frac{V(Y) - V(\mathbb{E}(Y|\mathbf{x}_{\sim i}))}{V(Y)} \tag{7}$$

The Sobol' indices can easily be obtained by Monte Carlo simulations, and a general R implementation is provided in Ref. [30]; chapter 8.

4.2. ALPERC

In this section we explain the developed AL procedure for Physical Experiments based on nonparametric Ranking and Clustering (ALPERC). In the text we focus on the rationale behind the method, while specific implementation details are reported in Algorithm 1.

The procedure is initiated by considering an initial dataset $\mathbf{\Omega}_0 = [\mathbf{D}_0\ \mathbf{Y}_0]$, where $\mathbf{D}_0$ with dimension $n_0 \times d$ is a MaxPro design with a limited number of levels for the factors [37], and $\mathbf{Y}_0$ with dimension $n_0 \times c$, is the matrix of responses, that are assumed to be independent. In this paper we opt for MaxPro designs with only discrete numeric factors for their properties in terms of good projection capabilities to all subspaces of the factors [38] and because they have been shown to work well in the setting of physical experiments when the data is analyzed with non-linear predictive models [39]. The adequate number of levels in $\mathbf{D}_0$ depends on the specific study and in general is the result of a trade-off between the experimental complexity and cost, and the need to provide enough granularity to the models to enable an appropriate estimation of the non-linear effects in the data. In another study [39] it has been shown that 6 levels for the factors may be sufficient for an estimation of the non-linearities in common physical phenomena. Then, another design $\mathbf{D}_{cand}$ including n_{cand} candidate experimental configurations is constructed following the same principles employed for $\mathbf{D}_0$. n_{cand} should be large enough to appropriately cover the design space, and should be significantly larger than the total number of experimental configurations that can be added throughout all the AL iterations. Nevertheless, the analyst should bear in mind that the computational effort (especially for the calculation of ranks) increases with n_{cand}, thus an option might be to temporarily augment $\mathbf{D}_0$ at each AL iteration by selecting a limited number of combinations (e.g. 100) from $\mathbf{D}_{cand}$ that minimize the MaxPro criterion [38,40], and to use this subset of $\mathbf{D}_{cand}$ as the candidate set at that AL iteration.

The next step consists in the construction of c predictive models $f_1(\cdot)$, …, $f_c(\cdot)$ that are trained on $\mathbf{\Omega}_0$ and on the estimation of the variable importance. In this paper we consider GP models and the use of total Sobol' indices (section 4.1) for a quantification of variable importance. The models are then used to obtain a quantification of prediction uncertainty of candidates in $\mathbf{D}_{cand}$.

At this point the nonparametric ranking procedure proposed by Ref. [41] is used. This method is based on the execution of nonparametric permutation tests that assume independent data and use the difference in means as test statistics. The permutation tests are applied for the comparison of some groups G that are the rows of $\mathbf{D}_{cand}$, and the dependent variable is the predictive uncertainty observed on the c responses. To ensure a proper sample size, we recommend a minimal value of $c = 3$. Once all the comparisons between the groups have been performed and the relevant p-values are computed, the ranking procedure described in Ref. [41] is applied and the vector $\mathbf{r}$ of ranking estimates is obtained. This provides a synthetic rank of candidates with respect to the uncertainty associated to all the responses [39]. The main advantage of this approach is that the rank assigns different positions only to those candidates whose global predictive uncertainty is significantly different after the execution of permutation tests, while a simple application of the ALM principle would not be able to do this.

Suppose that a region of the design space has the highest uncertainty, meaning that all the candidates from that area will rank the highest with respect to the uncertainty of predictions. If the AL procedure is solely driven by the predictive uncertainty related to each candidate, the algorithm would choose points that are close together in the uncertain region of the design space. Generally, this is inappropriate as one would prefer to explore different parts of the design space in order to reduce the predictive error globally. For this reason, a hierarchical agglomerative clustering algorithm [42] is employed to group candidates that are close together in the design space, where the similarity of two candidate configurations $\mathbf{x}$, $\mathbf{x}'$ is quantified using the weighted Euclidean distance:

$$\left(\sum_{i=1}^{d} w_i (x_i - x_i^{'})^2 \right)^{1/2} \tag{8}$$

where $\mathbf{w} = (w_1, \ldots, w_d)$ is the vector of variable importance, that is the vector of rescaled total sensitivity indices assigned to each variable in the models. The Silhouette index is chosen for the identification of the best number of clusters [43,44]. The additional correction provided by considering the weights in (8) ensures that if one specific variable is only scarcely relevant in all the models (i.e. the factor is inactive), the distance in that dimension is reduced, and candidates that differ with respect to "irrelevant" dimensions will be put into the same cluster. Similarly, if two candidates are spatially close but differ with respect to few important factors, their relative distance is increased, and they would tend to be assigned to different clusters.

At this point, a batch of candidate configurations to add in the next AL iteration must be selected. First of all, the size of the batch n_{add} must be defined, and we found that this depends on the specific application and is often chosen in conjunction with management. Some factors that can influence the choice of n_{add} are the number of test beds for parallel execution of the runs, or economic considerations in the case of trials requiring a stop to production. As a general rule, given the constraints imposed by economic factors, n_{add} should not be too large as this would affect the flexibility of the procedure. At this point, two different principles that guide the sequential selection of candidates can be chosen, one that favors *exploration* of the design space, and the other, more conservative one, favoring *exploitation* of existing knowledge.

Consider the case in which two candidates share the highest position in the rank (i.e. they are characterized by the same uncertainty) and belong to the same cluster (i.e. they come from the same weight-adjusted region in the design space). If the primary concern is the *exploration* of the design space, one should pick only one of these experimental configurations (the one with the highest mean uncertainty over the responses) and proceed by descending the rank until a new candidate is found that belongs to a different cluster and/or has a different position in the rank. On the contrary, if in the same situation *exploitation* is the main interest, one should perform some replicates of the configuration that has the highest mean uncertainty, while sharing the ranking position and cluster with others. In our experience, the number of replicates should be equal to the number of candidates that share the same ranking position and cluster, as this enables the possibility of sampling more in those areas of the design space that have the highest uncertainty. From a practical standpoint, better results are usually provided by opting for *exploration*, but the situation is different in the case of largely heteroscedastic systems. In these situations, the predictive models would benefit greatly from the execution of replicates [39] as they provide a clear view of noise not obfuscated by signal, that has been shown to benefit the prediction accuracy and computational efficiency of GP surrogate models [19]. Finally, n_{add} runs are selected from $\mathbf{D}_{cand}$ according to one of the principles explained above, and the analyst can proceed with the execution of the new trials. After the responses are collected, the new dataset $\mathbf{\Omega}_{add}$,

$_0$ can be concatenated to $\mathbf{\Omega}_0$ and the procedure can be iterated n_{iter} times, i.e. until a certain level of accuracy of the models is reached, or the budget allocated to the experiment is terminated.

Algorithm 1. ALPERC

Require: n_0, n_{cand}, n_{add}, n_{iter}, `exploration`, `exploitation`
for $j = 0 : n_{iter}$ **do**
  if $j = 0$ **then**
    Generate initial design $\mathbf{D}_0^{n_0 \times d}$
    Collect responses $\mathbf{Y}_0^{n_0 \times c}$ and rescale each to $[0, 1]$
    Generate design of candidate points $\mathbf{D}_{cand}^{n_{cand} \times d}$
    $\mathbf{\Omega}_j \leftarrow \mathbf{\Omega}_0 = [\mathbf{D}_0\ \mathbf{Y}_0]$
  else
    Remove $\mathbf{D}_{add,j-1}$ from $\mathbf{D}_{cand}$: $\mathbf{D}_{cand} = \mathbf{D}_{cand} \setminus \mathbf{D}_{add,j-1}$
    $\mathbf{\Omega}_j = \begin{bmatrix} \mathbf{\Omega}_{j-1} \\ \mathbf{\Omega}_{add,j-1} \end{bmatrix}$, where each column of $\mathbf{Y}_j$ is rescaled to $[0, 1]$
  end if
  for $i = 1 : c$ **do**
    Fit model $f_{j,i}(\mathbf{\Omega}_j) = \hat{f}_{j,i}(\cdot)$
    Compute importance and rescale it to $[0, 1]$: $\mathbf{S}_{T,i}|\hat{f}_{j,i}(\cdot) = (S_{T,i,1}, \ldots, S_{T,i,d})$
    Get uncertainty of candidate points $\boldsymbol{\sigma}_{cand,i}|\hat{f}_{j,i}(\mathbf{D}_{cand}) = (\sigma_{i,1}, \ldots, \sigma_{i,n_{cand}})$
  end for
  Save $\mathbf{S}_T = \{\mathbf{S}_{T,1}, \ldots, \mathbf{S}_{T,c}\}$
  Save $\boldsymbol{\sigma}_{cand} = \{\boldsymbol{\sigma}_{cand,1}, \ldots, \boldsymbol{\sigma}_{cand,c}\}$
  Get $\mathbf{r}$, rank of candidates in $\mathbf{D}_{cand}$ with respect to $\boldsymbol{\sigma}_{cand}$
  Get vector $\mathbf{h}$ of clusters of candidates in $\mathbf{D}_{cand}$, using $\left(\sum_{k=1}^{d} w_k (x_k - x'_k)^2\right)^{1/2}$ as similarity measure, where $w_k = \frac{\sum_{i=1}^{c} S_{T,i,k}}{c}$
  Construct matrix $\mathbf{\Lambda} = [\mathbf{D}_{cand}\ \mathbf{r}\ \mathbf{h}]$, arranged with respect to $\mathbf{r}$
  if `exploration` **then** ▷ no replicates
    for $i = 2 : n_{cand}$ **do**
      if $r_i = r_{i-1}$ and $h_i = h_{i-1}$ **then**
        if $\bar{\sigma}_{cand,i} \leq \bar{\sigma}_{cand,i-1}$ **then**
          remove $\mathbf{\Lambda}_i$
        else
          remove $\mathbf{\Lambda}_{i-1}$
        end if
      end if
    end for
    Generate $\mathbf{D}_{add,j}$ including the first n_{add} rows and d columns of $\mathbf{\Lambda}$
  end if
  if `exploitation` **then** ▷ do replicates
    $n_{repl,1} \leftarrow 1$
    for $i = 2 : n_{cand}$ **do**
      $n_{repl,i} \leftarrow 1$
      if $r_i = r_{i-1}$ and $h_i = h_{i-1}$ **then**
        if $\bar{\sigma}_{cand,i} \leq \bar{\sigma}_{cand,i-1}$ **then**
          remove $\mathbf{\Lambda}_i$
          $n_{repl,i-1} \leftarrow n_{repl,i-1} + 1$
        else
          remove $\mathbf{\Lambda}_{i-1}$
          $n_{repl,i} \leftarrow n_{repl,i-1} + 1$
        end if
      end if
    end for
    Replicate each $\mathbf{\Lambda}_i$ $n_{repl,i}$ times and include in $\mathbf{D}_{add,j}$ the first n_{add} rows and d columns of $\mathbf{\Lambda}$
  end if
  Collect responses $\mathbf{Y}_{add,j}^{n_{add} \times c}$
  $\mathbf{\Omega}_{add,j} = [\mathbf{D}_{add,j}\ \mathbf{Y}_{add,j}]$
end for

5. Illustrative example

In this section we use an illustrative 2D example to provide an intuitive explanation of the functioning of ALPERC. We consider four nonlinear test functions:

$$\begin{aligned}
&\text{f1:} && 2\sin A^2 - \exp B^2, \quad \text{with } A \in [0, 2],\ B \in [0, 1] \\
&\text{f2:} && \log A^2 - 5\sin B^3, \quad \text{with } A \in [0.15, 1],\ B \in [0.4, 2] \\
&\text{f3:} && \sin\frac{1}{A} + 3\cos B, \quad \text{with } A \in [0.2, 1],\ B \in [0.5, 1.5] \\
&\text{f4:} && 2\exp A + \frac{1}{2}B^3 + AB, \quad \text{with } A \in [-2, 0.5],\ B \in [-1.3, 0]
\end{aligned} \tag{9}$$

and we use ALPERC to drive the sequential selection of points with the aim of building accurate emulators of the responses. We provide surface plots of each rescaled function in Fig. 2, and to better represent a physical setting, we add to each function an error component $\varepsilon \sim \mathcal{N}(0, \sigma^2_{het})$ that increases linearly with the response. That is, at a specific design configuration $\mathbf{x}$: $\sigma_{het} = \alpha(f(\mathbf{x}) + \gamma)$, with $\alpha = (\sigma_{y_i} - 0.05\sigma_{y_i})/(\max y_i - \min y_i)$, $\gamma = (0.05\sigma_{y_i} - \alpha \min y_i)/\alpha$, where $\max y_i$ and $\min y_i$ with $i = 1, \ldots, 4$ are the maximum and minimum of each function, and σ_{y_i} is the standard deviation quantifying the variability of each function. We initiate the procedure with a MaxPro discrete design $\mathbf{D}_0$ of size 20, with 6 levels for each factor, and we collect the noisy realizations. We then fit a heteroscedastic GP [20] to each response. In this case, due to the simplicity of the situation and low dimensionality of the problem, we do not compute the variable importance. At the initialization of the procedure, we build a MaxPro design $\mathbf{D}_{cand}$ with 200 data configurations, and at each iteration we select 100 combinations that minimize the MaxPro criterion as candidate points, as described in section 4.2. We set $n_{add} = 5$, and we proceed by sequentially adding n_{add} configurations for 10 iterations. In this case, we opt for the *exploration* approach, but since the candidate points at each iteration (100) are significantly more than the number of

factorial combinations (6^2), some replicates will be included at subsequent iterations. We believe that in an illustrative example this can help in the visualization of the AL procedure, as we expect a good method to sample several points in uncertain regions of the design space. Nevertheless, we emphasize that this approach is not usually feasible in practical applications as the number of factorial combinations is usually significantly larger than the number of runs that can be afforded for the study, and in practical situations the *exploitation* approach must be followed for inclusion of replicates in the procedure.

Fig. 3 shows the updates of the AL algorithm. Each panel reports the contour plot of the response surface obtained as the mean of the rescaled 2D functions from (9), since the final objective is a reliable prediction of all the functions altogether. The first panel shows the design $\mathbf{D}_0$ (crosses) and the cumulative number of samples taken from each design configuration throughout all the AL iterations. Each plot is then specific to each iteration and shows the candidate points grouped in different clusters, according to the symbols. The ranking of each candidate point according to the predictive uncertainty of the responses is presented via the color gradient associated to each symbol, from white (low position, small uncertainty) to dark blue (high position, high uncertainty). There are several noteworthy insights. At *iteration* $= 0$, the algorithm favors those configurations that lie at the boundary of the design space, that is a well-known property of variance-based AL strategies, since uncertainty tends to be higher at the frontiers of the design. At this iteration the configurations are selected from three different clusters, and we see how only one configuration is chosen in case several candidates share the same position in the rank and belong to the same cluster (e.g. configurations (0.8, 1) and (1, 0.8)). Already from *iteration* $= 1$ the algorithm has identified the high variance regions in the design space, since from this iteration all the configurations added lie in the yellow area of the plots. From *iteration* $= 2$ the algorithm starts to focus on central areas of the design space, especially on the lower-right region, where uncertainty is the highest. The algorithm then proceeds by selecting configurations in the informative area of the design space up until the last iteration. Furthermore, from *iteration* $= 5$ the candidate points no longer completely cover the central region of the design, indicating that at the specific iteration those points are not among the 100 configurations from $\mathbf{D}_{cand}$ that minimize the MaxPro criterion. Throughout all the iterations,

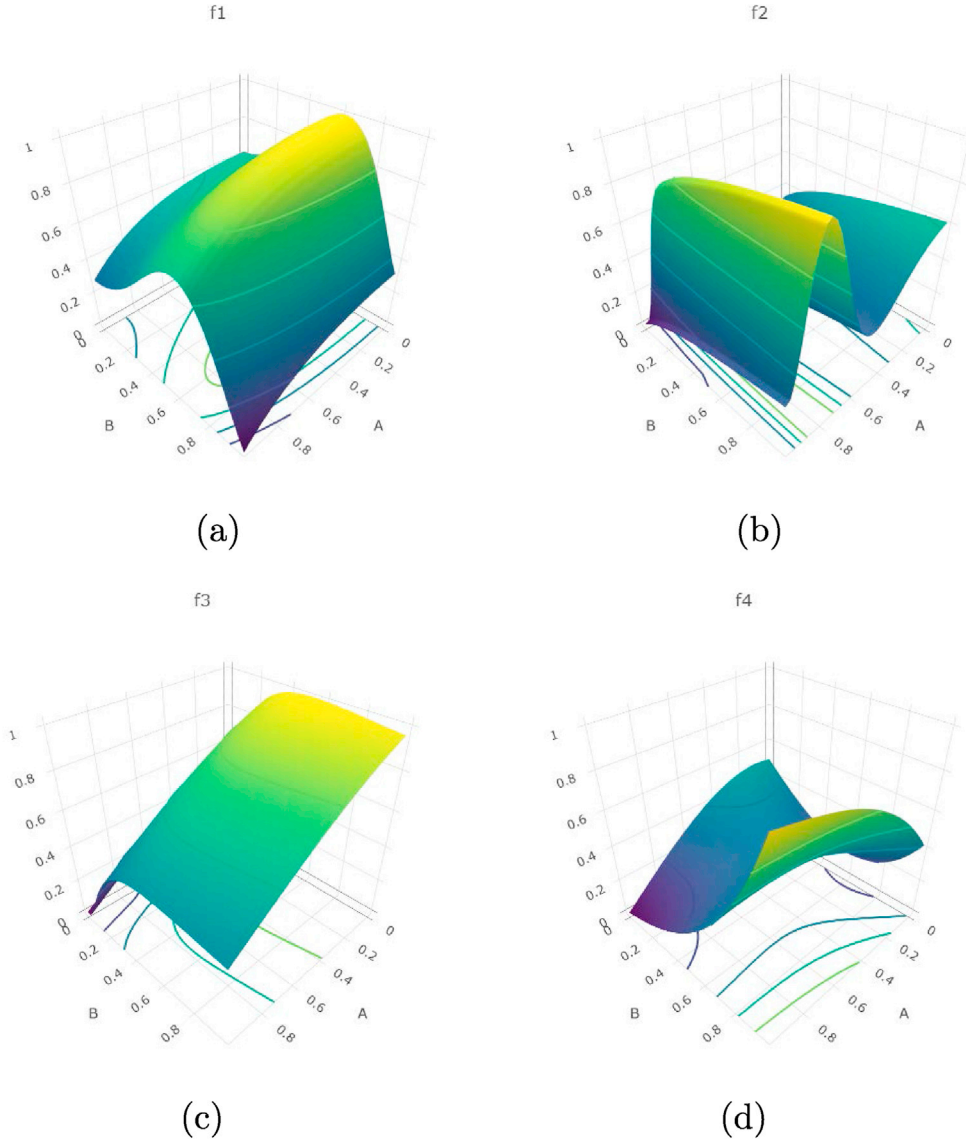


Fig. 2. The four 2D test functions.

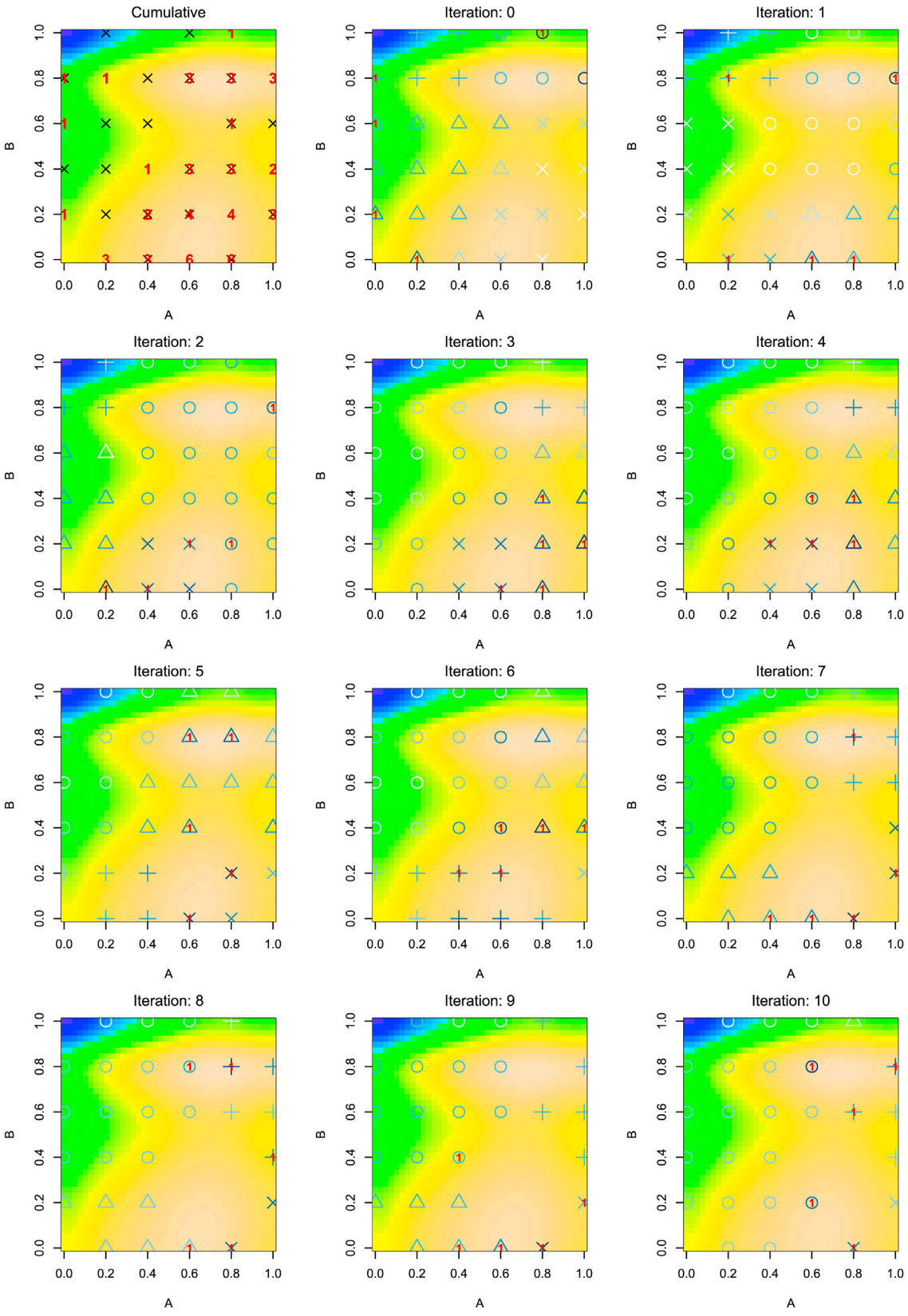
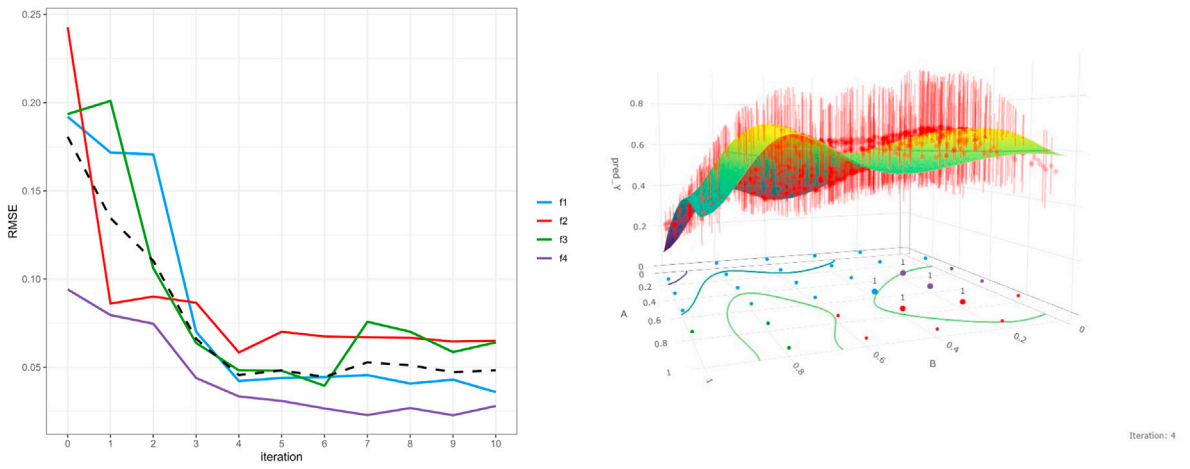


Fig. 3. Depiction of ALPERC applied on the 2D illustrative example. The first panel shows the initial design (crosses) and cumulative points selected throughout the 10 AL iterations (numbers). The other panels focus on specific AL iterations, showing the candidate points grouped in clusters (symbols) and position in the rank (color). The configurations selected at each iteration are indicated in red. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

we see how ALPERC samples more in the areas with the highest uncertainty, whilst ensuring a proper exploration of the design space.

Fig. 4-a shows the test error computed on a noiseless 1000-point random LHD, indicating how this steeply decreases at initial iterations



(a) (b)

Fig. 4. The test error (for each 2D function and the average) throughout the 10 AL iterations (a), and the average predictions on the test set at *iteration* = 4 (b).

Table 1
Summary of the test functions for the simulation study.

Name	Description	# dimensions
Borehole	models water flow through a borehole	6
OTL circuit	models an output transformerless push-pull circuit	6
Piston	models the circular motion of a piston within a cylinder	6
Piston Mod	a modification of Piston function, with increased non-linearity	6
Robot arm	models the position of a robot arm which has 3 segments	6
Rosenbrock Function	is a popular test problem for optimization algorithms	6
Wing weight	models a light aircraft wing	6

Table 2
Summary of the noise structures for the simulation study.

Noise σ	Type	Description
$\sigma_{hom} = 0.2\sigma_{y_i}$	Homoscedastic	20% noise
$\sigma_{hom} = 0.5\sigma_{y_i}$	Homoscedastic	50% noise
$\sigma_{het,min} = 0.05\sigma_{y_i}$, $\sigma_{het,max} = 1\sigma_{y_i}$	Heteroscedastic	5% noise at min y_i and 100% noise at max y_i (low5_high100)
$\sigma_{het,min} = 0.05\sigma_{y_i}$, $\sigma_{het,max} = 5\sigma_{y_i}$	Heteroscedastic	5% noise at min y_i and 500% noise at max y_i (low5_high500)

and reaches a plateau at *iteration* = 4. Fig. 4-b shows the noiseless surface (obtained as the mean of the rescaled 2D functions) together with the point predictions and corresponding predictive uncertainty (obtained as the average of the predictions and uncertainties of all 2D functions). For visualization purposes, the error bars are shown only on 500 points of the test data. From the figure it can be seen how the predicted points closely follow the true surface and the uncertainty reflects the noise structure, which is smaller where the mean response is lower. Furthermore, the predictive uncertainty has the typical football-shape of GP models [30] as it is smaller in correspondence with design points and increases as it gets farther away (e.g. B in [0.4,0.6]).

6. Simulation study

In this section we compare the predictive accuracy of ALPERC against competitive methods from the literature. To provide a comprehensive comparison, we carry out an extensive simulation study considering several test functions, noise settings, and two levels of sparsity: no sparsity and 25% sparsity, where sparsity refers to the number of inactive effects in an experiment.

6.1. Simulation setting

We consider a simulation protocol similar to the one presented in Ref. [39]. We choose 7 test functions (Table 1) from the literature, mainly emulating physical processes [45], restricted to having 6 active dimensions as described in the supplemental material provided in Ref. [39]. Each dependent variable was standardized and the independent variables were normalized to [0–1], as described in Ref. [39].

We examine both homoscedastic and heteroscedastic noise cases. For the homoscedastic case, we consider a random error component $\varepsilon \sim \mathcal{N}(0, \sigma^2_{hom})$, with σ_{hom} either $0.2\sigma_{y_i}$ or $0.5\sigma_{y_i}$, where σ_{y_i} with $i = 1, \ldots, 7$ is the standard deviation associated to each test function, estimated from 100 random LHDs with 500000 points each [39]. In the heteroscedastic noise case, we consider a random noise component $\varepsilon \sim \mathcal{N}(0, \sigma^2_{het})$, where σ_{het} increases linearly with the response, that is at a design configuration $\mathbf{x}$: $\sigma_{het} = \alpha(f(\mathbf{x}) + \gamma)$, with $\alpha = (\mathtt{m}\sigma_{y_i} - 0.05\sigma_{y_i})/(\max y_i - \min y_i)$, $\gamma = (0.05\sigma_{y_i} - \alpha \min y_i)/\alpha$, and $\mathtt{m}$ is either 1 or 5. A summary of the noise settings is provided in Table 2. For further details and a visualization of the impact of noise on the test functions, please refer to the work of Ref. [39].

The levels of sparsity considered are: no sparsity, in the case of the experiment which considers 6 factors, all of which are active, and 25% sparsity, in the case of 2 inactive factors being added to the 6 active ones. The inactive factors are obtained as random permutations of the other factors, and do not determine the outcome of the test functions but are included as predictors in the models.

The performance of ALPERC and competitive methods is quantified by means of their mean predictive accuracy over the 7 test functions at each AL iteration. In particular, to obtain a synthetic comparison of the different methodologies, the nonparametric ranking algorithm developed by Ref. [41] is applied to analyze the global mean predictive error over all the AL iterations obtained with the different AL methodologies. At each iteration, the predictive accuracy is measured as the mean root mean squared error (RMSE) obtained on a test dataset over the 7 test functions. With the aim of comparing the predicted values over the entire design space, we generate the true noiseless realizations of all the test functions on a 1000-point random LHD, and we consider the resulting matrix as the test dataset. Therefore, in short, at each iteration we employ the test dataset to compare the predictions provided by models that are trained on few noisy data points to the true noiseless realizations of the test functions. For each noise setting, level of sparsity, and AL method

considered, we repeat the simulation over 5 random initializations of the procedure.

6.2. ALPERC implementation and competitive methods

At the initialization of each simulation run we set the ALPERC parameters from Algorithm 1 as $n_0 = 52$ [39], $n_{cand} = 500$ (with selection of 100 points at each iteration, as explained in section 4.2), $n_{add} = 10$, and $n_{iter} = 10$, and we focus on the homoscedastic and heteroscedastic GP models with anisotropic Gaussian kernel [32]. The same initial parameters are considered for the competitive methods that we will introduce shortly. Furthermore, we also test a variant of the ALPERC algorithm in which the vector of weights for construction of clusters is $\mathbf{w} = \mathbf{1}$, that is the grouping of candidate configurations into clusters is not adjusted by the importance associated to each dimension (ALPERC_unw). The design $\mathbf{D}_0$ is unreplicated, and replicates are included in the sequential AL steps when the *exploitation* approach is adopted. Preliminary results not reported here show that the *exploitation* strategy is to be preferred solely in the presence of severe heteroscedasticity, thus ALPERC is implemented with the *exploration* approach in all noise settings except the one with the highest heteroscedasticity. This is also in line with the findings from Ref. [39].

The competitive methods for sequential selection of design points that we use are: an augmentation based on minimization of the MaxPro criterion (MaxPro_aug), an augmentation based on D-optimality (D_opt), sequential selection based on maximum variance of predictions (ALM), sequential selection based on the expected variance reduction over the input space (ALC), and a non-sequential approach in which at each iteration the models are retrained on a design of appropriate size (NO_AL). All the methods are compliant with the requirements listed in section 3, also with minor adaptations as specified below.

The strategies based on an augmentation following the MaxPro criterion and D-optimality do not make use of the information provided by the predictive models at each iteration, thus are more similar to traditional sequential DOE approaches than to AL strategies where the learner is in control of the data from which it learns [1,46]. The approach based on the augmentation of the existing MaxPro design is implemented in Ref. [40] and consists in a selection from $\mathbf{D}_{cand}$ of the n_{add} configurations that optimize the MaxPro criterion, considering the points already included in the design. The sequential D-optimality approach is implemented using the functions from Ref. [47], and aims to obtain a precise estimate of the model parameters by sequentially sampling n_{add} candidates from $\mathbf{D}_{cand}$. In a multivariate setting it is conceptually equivalent to the DOWAL algorithm proposed by Ref. [11].

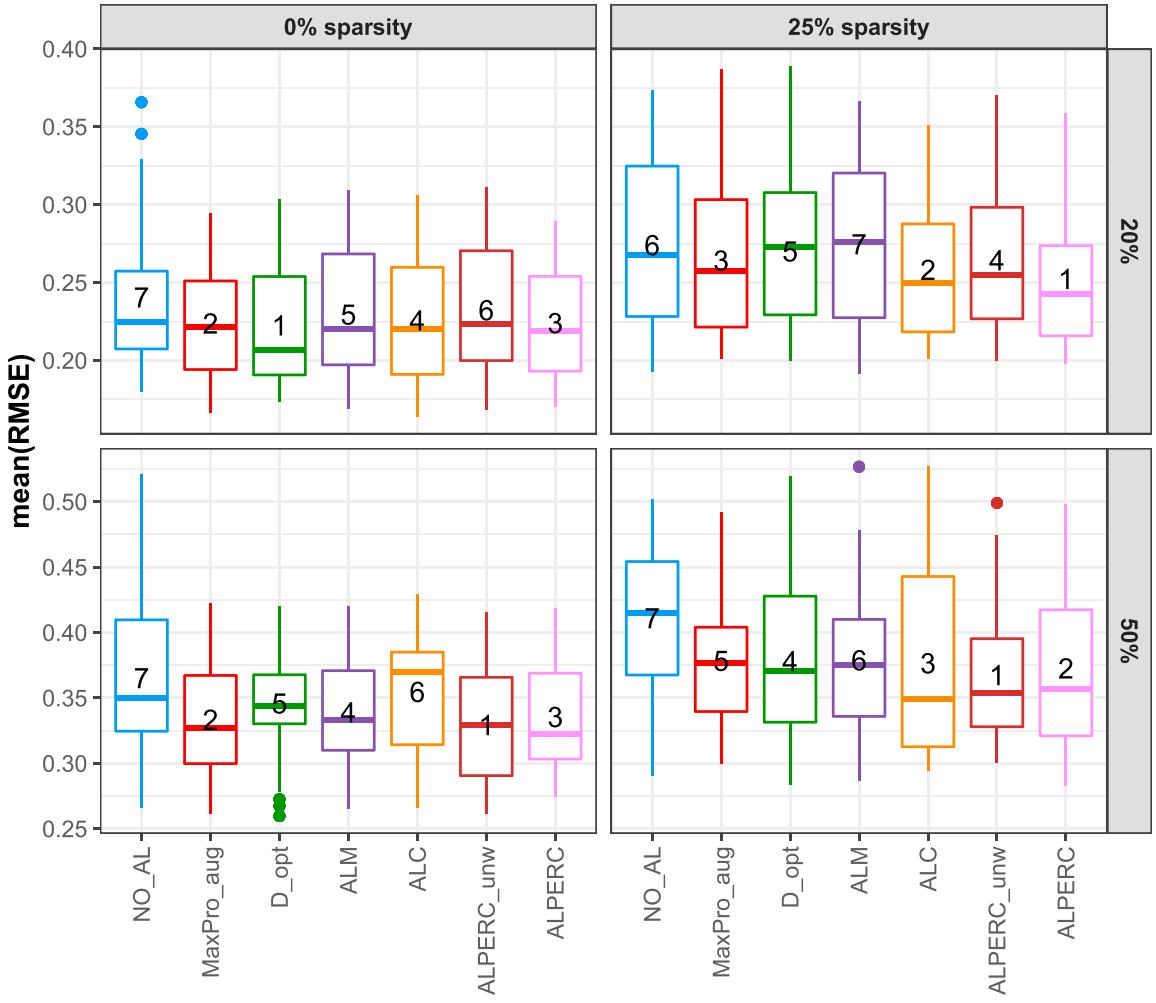


Fig. 5. Synthetic simulation results for the homoscedastic noises. The boxplots are obtained by taking into consideration the mean RMSE realizations from the AL iterations 1 to 10. The numerals reside in the location of the mean, and indicate the relative arithmetic rank of the mean of that group.

The sequential selection based on the maximum variance of predictions is analogous to ALM, and in a multivariate experiment is the VWAL algorithm from Ref. [11]. One difference is that instead of adding one configuration at a time, in this case we add the n_{add} configurations that have the highest mean predictive uncertainty over the test functions. AL based on the expected variance reduction over the input space is the ALC criterion, and in this paper we use the implementation for anisotropic Gaussian GPs included in Ref. [48] by adding at each iteration the n_{add} configurations that have the highest mean ALC value. In order to limit the impact of using two different R libraries to train the GPs (given that the ALC strategy is implemented using the laGP package, while in all other cases we employ hetGP), we always use the θ and g estimated by hetGP (in the homoscedastic or heteroscedastic versions depending on the noise setting). However, the heteroscedastic GP is not implemented in laGP therefore only the homoscedastic model is used for ALC.

The last method we use for comparison is the case in which we do not sequentially add configurations in each iteration, but rather we build a new MaxPro design with discrete numeric factors that has the same number of configurations as the AL counterparts at the same iteration. This provides a comparison of the performance of non-sequential methods with the same budget in terms of experimental runs, and shows what analysts should expect if they choose to proceed in a non-sequential manner.

6.3. Simulation results and discussion

Figure A.12 (appendix) shows the test error computed for all the AL strategies at each sequential step and a 20% homoscedastic noise level. The results are not displayed at *iteration* = 0 as they are the same for all the methods since given a noise setting, $\mathbf{D}_0$ is the same. In general, and as expected, there is a progressive decrease in the error as more data is added to the models. However, the decrease in test error is not the same for all the AL strategies, and at 25% sparsity in particular the test error may in fact increase at subsequent iterations for some methods.

Figs. 5 and 6 provide a synthetic visualization of the global results across iterations 1 to 10, considering all the noise settings and sparsity levels. These synthetic visualizations are easier to interpret than Figure A.12, and can offer some interesting insights. For the homoscedastic noises (Fig. 5) we notice how performance of the different methodologies depends on both the noise magnitude and sparsity level. For low noise and 0% sparsity, many competitive methods perform more or less equally, and both D_opt and MaxPro_aug strategies perform as well as or better than the other AL strategies. This may be rather surprising given they are simple approaches when compared to the more sophisticated AL methodologies. However, it must be emphasized that for low noise and 0% sparsity, the insights provided by the models may not be crucial to obtain accurate predictions as there is no particular need to identify high

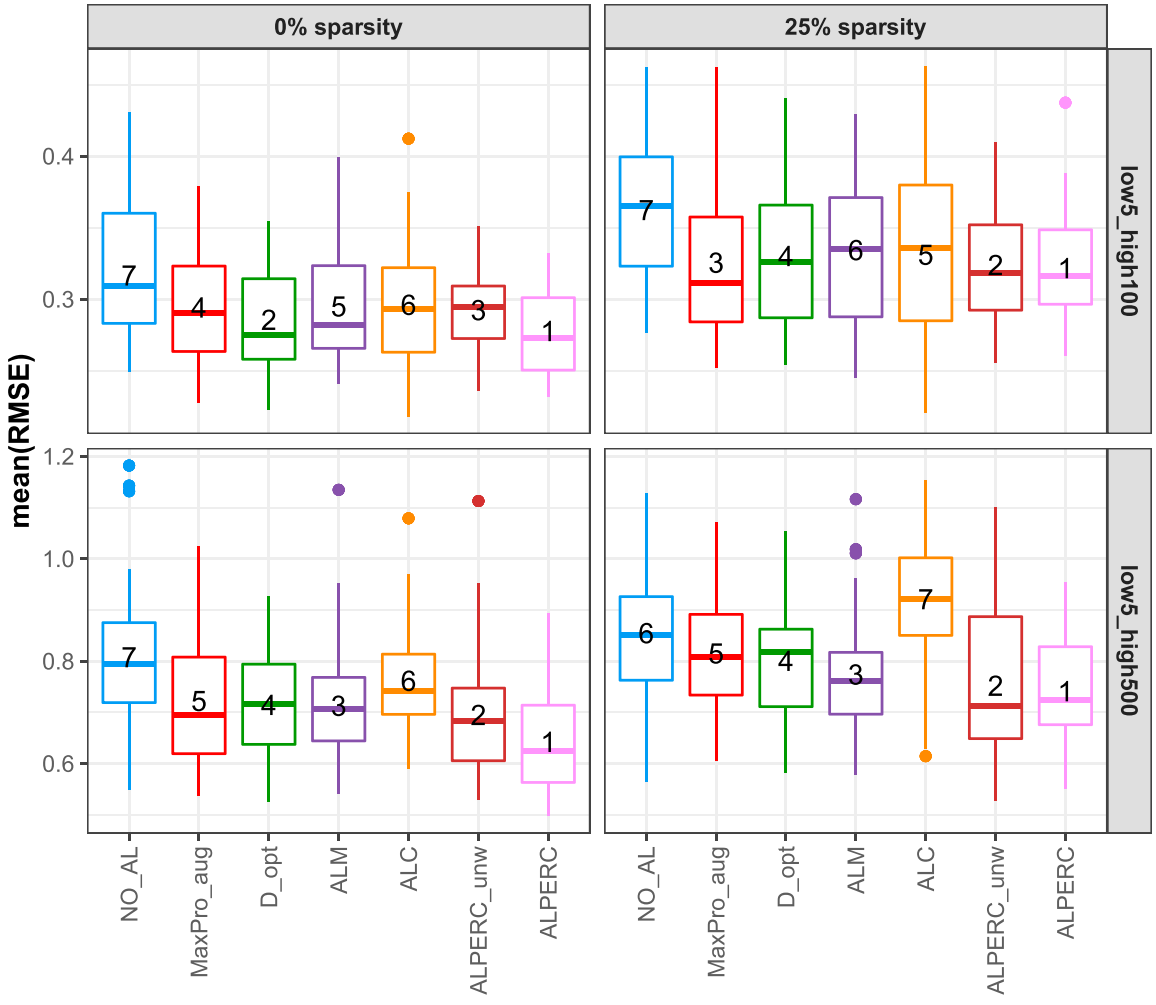


Fig. 6. Synthetic simulation results for the heteroscedastic noises. The boxplots are obtained by taking into consideration the mean RMSE realizations from the AL iterations 1 to 10. The numerals reside in the location of the mean, and indicate the relative arithmetic rank of the mean of that group.

Table 3
Simulation study: synthetic rank of the AL methods for the homoscedastic noise cases. Low rank values indicate small mean predictive error.

Method	0% sparsity		25% sparsity	
	20%	50%	20%	50%
NO_AL	7	7	6	7
MaxPro_aug	**1**	**1**	3	5
D_opt	**1**	5	5	5
ALM	5	4	6	2
ALC	4	6	2	2
ALPERC_unw	6	**1**	4	**1**
ALPERC	**1**	**1**	**1**	2

Table 4
Simulation study: synthetic rank of the AL methods for the heteroscedastic noise cases. Low rank values indicate small mean predictive error.

Method	0% sparsity		25% sparsity	
	low5_high 100	low5_high 500	low5_high 100	low5_high 500
NO_AL	7	7	7	6
MaxPro_aug	2	2	2	5
D_opt	2	2	2	4
ALM	2	2	6	3
ALC	2	6	2	7
ALPERC_unw	2	2	2	**1**
ALPERC	**1**	**1**	**1**	**1**

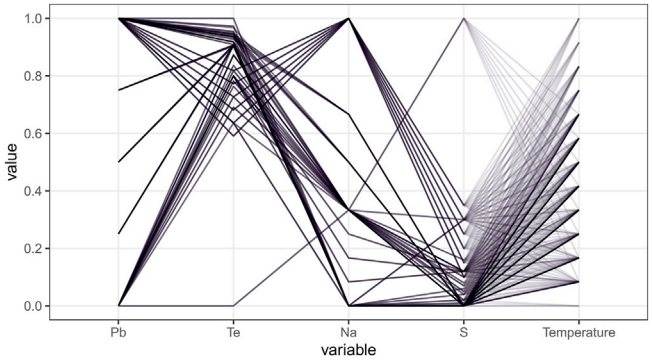


Fig. 7. Parallel coordinates plot of predictors in the thermoelectric dataset, after rescaling to [0–1]. The data is undesigned, as samples do not appropriately cover the design space.

uncertainty regions in the design space and an appropriate spatial location of the new experimental configurations with respect to the existing ones may already be sufficient. As the noise and (especially) sparsity increase, we notice how the sophisticated AL strategies provide an advantage. Table 3 reports a ranking of the AL strategies using the data from Fig. 5, employing the nonparametric ranking procedure of [41]; and assuming paired data and $\alpha = 5\%$. This rank provides inferential information regarding the performance of the different AL strategies, while only descriptive information can be obtained from Fig. 5. The results show that ALPERC is the best performer in all situations except in the case of 50% noise and 25% sparsity, when ALPERC_unw is the best performer. In the case of 0% sparsity, the MaxPro_aug method performs as well as ALPERC, but when some factors are inactive, none of the competitors considered, including ALM and ALC, performs as well as or better than either ALPERC (at 20% noise) or ALPERC_unw (at 50% noise).

The advantages of ALPERC methods are even more evident if the heteroscedastic noise settings are considered because in this context ALPERC provides the lowest global mean predictive error over all the iterations considering all noise intensities and sparsity levels (Fig. 6), followed by ALPERC_unw in all cases except the one with the lowest noise and 0% sparsity. As heteroscedasticity becomes more severe, the performance improvement of ALPERC is even more stark. It should be noted that at the highest level of heteroscedasticity, ALPERC and ALPERC_unw are the only methods that include replicates (*exploitation* approach). However, results not reported here show that if at each AL iteration the candidate configuration with the highest mean predictive uncertainty selected by ALM (the closest competitor to ALPERC methods in this situation) is replicated 3 times to approximately match the level of replication of ALPERC, the resulting predictive performance is underwhelming, and the "replicated ALM" approach performs worse than ALC and only marginally better than the non-sequential strategy (at 0% sparsity). Therefore, if we consider the more extreme heteroscedastic setting, the advantage of ALPERC does not solely depend on the inclusion of replicates, but is attributable to the core principles driving the algorithm.

The ranks in Table 4 confirm the descriptive results from Fig. 6 and show that ALPERC significantly ($\alpha = 5\%$) outperforms all the competitors in all the input-dependent noise situations; ALPERC_unw always ranks second, apart from one case (low5_high500, 25% sparsity) when it matches the performance of ALPERC. Among the AL competitors it is worth mentioning that ALC performs very poorly in the presence of severe heteroscedasticity, and this might be in part attributable to the use of the homoscedastic GP as already explained.

From these simulations we can conclude that ALPERC significantly

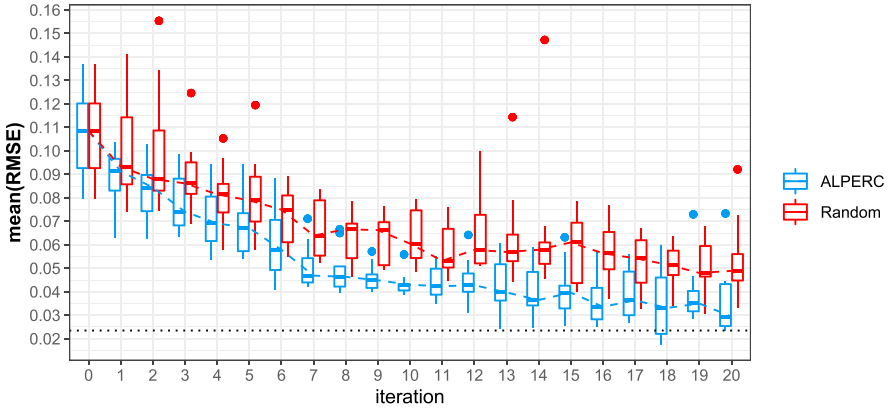


Fig. 8. Case study results of the average RMSE at each AL iteration. The dotted line corresponds to the median mean(RMSE) when all training data is used for training the models.

outperforms its competitors in most situations. In only one case ALPERC is not the best performer (50% homoscedastic noise, 25% sparsity), and it is beaten by its ALPERC_unw counterpart, i.e. ALPERC when the clusters are built considering the unweighted Euclidean distance. All the other competitors from the literature perform worse than or, at best, as well as ALPERC, although this happens only in some situations when the contribution of adaptively sampling the experimental configurations using insights from the models is already expected to be marginal. The advantages of ALPERC are even more pronounced when heteroscedasticity increases. It is evident that as the noise structure becomes more complex, the selection of points based on an identification of the high-variance regions in the design space becomes more important. However, this alone may not be enough, as proved by the performance of ALM that employs a simple approach for the selection of candidates with high uncertainty of predictions. The twofold feature of ALPERC, namely its ranking points according to predictive variance while at the same time taking the relative spatial location of candidates into account, shows the most gains in this situation. Furthermore, adjusting the attribution of candidates to clusters with respect to the relative importance of each variable in the models generally also provides a slight but significant improvement in performances.

A final point worthy of note is the disappointing performance of the non-sequential strategy (NO_AL). This method ranks last in nearly all situations, systematically providing worse predictions than the sequential approaches, despite the models being trained on the same number of data points. Therefore, in general, worse results should be expected if the choice is made to proceed with a DOE in which all the runs are carried out in a single batch and no follow up trials are planned. Given limitations in terms of experimental runs, a better choice is to proceed sequentially by executing an initial small DOE and then iteratively adding configurations following an AL principle. Although the importance of sequential experimentation is recognized in the DOE community [24], non-sequential approaches are still largely preferred by practitioners, even when ML models are employed for analysis [46]. This is in contrast to the intrinsic *learning* nature of ML models, and the results of this simulation confirm that an approach in which the predictions of the models guide data collection in the next iteration is to be preferred.

7. Case study: thermoelectric materials

7.1. Overview

Thermoelectric materials are solid-state energy converters that turn thermal energy into electrical energy. The thermoelectric properties are crucial for many applications including heating, cooling, temperature control, and polymerase chain reaction systems for DNA replication among others [49]. Recently, thermoelectric materials have been garnering attention for use within sustainable power generation systems as they can convert waste heat (e.g. from incinerators or engines) into electrical energy [50]. The efficiency of thermoelectric conversion is governed by the performance index ZT [49]:

$$ZT = \frac{Sb^2 \sigma T}{\kappa} \tag{10}$$

where κ, Sb, σ and T are thermal conductivity, Seebeck coefficient, electrical conductivity and operating temperature, respectively. In this study we employ ALPERC to develop models to predict the thermal conductivity, Seebeck coefficient and electrical conductivity of thermoelectric materials given the composition and operating temperature.

7.2. Experimental data and ALPERC implementation

The dataset employed for this study was retrieved from the *Starrydata2* web system, that collects experimental data on thermoelectric materials from published papers [51]. The samples were collected at an operating temperature in the range [300 K–900 K], and the composition included Pb, Te, Na and S. In total, the dataset consists of 457 samples. Predictors and responses were rescaled to [0–1] prior to model fitting. Since the data comes from several published papers, the dataset is undesigned (Fig. 7).

The thermoelectric dataset is randomly split into training (80%) and test (20%) sets. For robustness, the splitting procedure was repeated 10

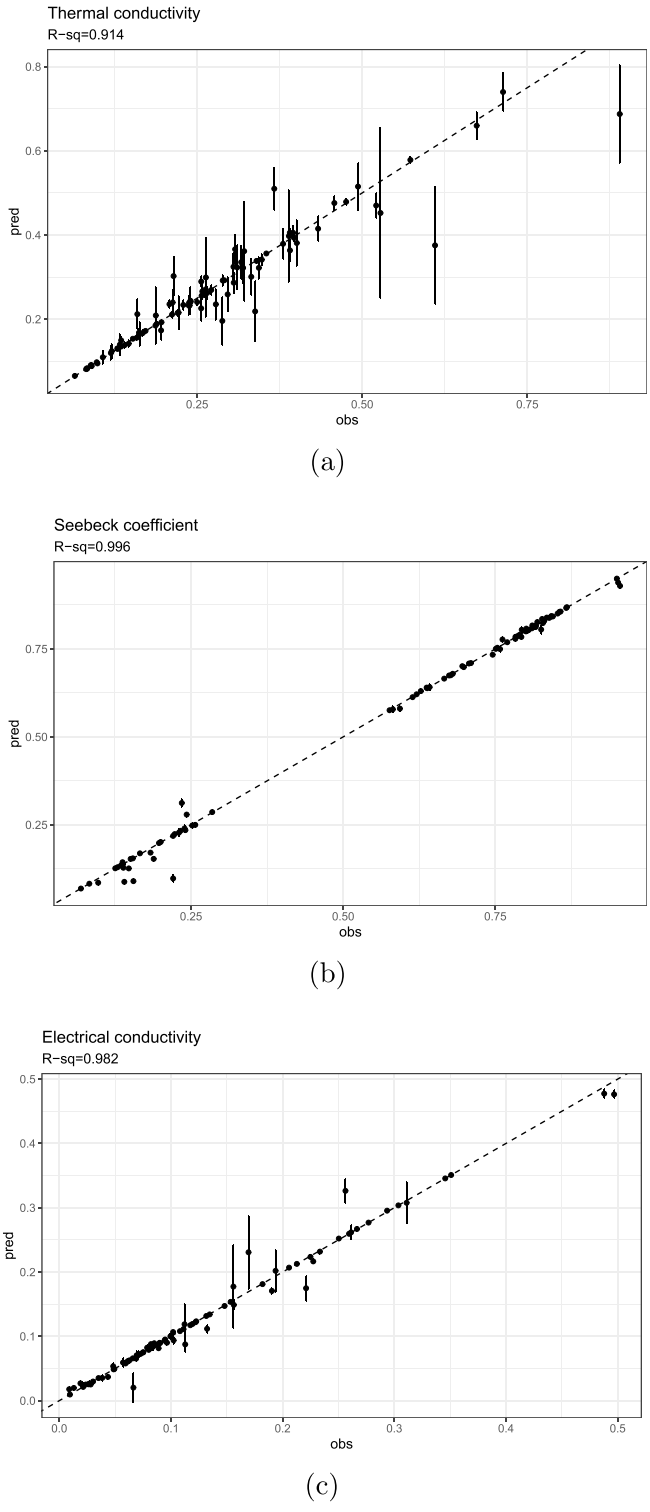


Fig. 9. Case study scatter-plots of observed by predicted values on the test data, considering the training-test partition that leads to the best results at *iteration* = 20 for ALPERC.

times, and ALPERC was implemented in each case. On each occasion the initial parameters of ALPERC were: $n_0 = 40$, $n_{cand} = 100$, $n_{add} = 10$, $n_{iter} = 20$ and the *exploration* approach was followed. $\mathbf{D}_0$ was randomly sampled from the training set, and at each AL iteration, $\mathbf{D}_{cand}$ was built including the n_{cand} configurations that minimize the MaxPro criterion taking into account the trials already included in the design. GP models with Matérn $\nu = 5/2$ covariance function [28] and total Sobol' indices were employed as predictive models and for quantification of variable importance, respectively.

7.3. Case study results and discussion

Fig. 8 shows the evolution of the mean test error for ALPERC compared with a random selection of candidates at each AL iteration. The figure also reports the median mean(RMSE) when all training data is employed, that is more than a 50% increase in data size if compared with ALPERC at *iteration* = 20. Two results are evident: ALPERC provides substantial performance improvements over a random candidate selection from the training data, and comparable results are obtained by ALPERC and models trained on the whole training data while employing substantially less resources. The goodness of the models at the last AL iteration is also proved by the "observed by predicted" scatter-plots in Fig. 9, that highlight optimal predictive accuracy for all the models, also in terms of R^2 calculated on the test data. Uncertainty quantification is also appropriate, as the error bars are much narrower near the 45° line, and increase as points move farther away.

Fig. 10 highlights another property of the ALPERC algorithm. When the procedure is initiated, none of the configurations that have Pb = 0, Te = 0 and S = 1 are included in the design. Consequently, these configurations are among those with the highest uncertainty in $\mathbf{D}_{cand}$, since they are very different to any combination included in $\mathbf{D}_0$. Therefore, at *iteration* = 0, ALPERC indicates the inclusion of some of these configurations in the design. Note, however, that only two such configurations are added at this step in order not to dedicate all efforts to the exploration of a single region of the design space, but rather to divide efforts between the exploration of several areas with high uncertainty. Then, when the models are trained at *iteration* = 1 and residual high uncertainty is detected for samples with Pb = 0, Te = 0 and S = 1, another of these configurations is added to the design. At subsequent iterations, more of these samples are included in the training data, as shown by the results at the last iteration (Fig. 10-d).

Fig. 11 displays the evolution of the variable importance over ALPERC iterations. This provides insights into the rationale driving the GP models, and displays the progressive *learning* of the models as more data is added. For each iteration, the points indicate in what proportion each predictor contributes to the response, both in terms of main effect and interaction with other variables. This quantity is used in ALPERC to adjust the attribution of each candidate to a cluster. More detailed information can be provided by calculating the difference between total sensitivity indices and first-order indices (not computed here), thus distinguishing what part of variability in the response can be attributed to interactions between variables and what part to main effects [52]. Fig. 11 shows that there is some variability in the variable importance throughout the iterations. While this is in part a consequence of the model's progressive learning, it is also attributable to the shape of the data. For instance, consider the case of "electrical conductivity" (Fig. 11-c): at initial iterations Temperature is the most important predictor, but it becomes less relevant as AL progresses. This might be because more levels of Temperature are included in the data compared to the other factors (Fig. 7), therefore the models are easily tricked into

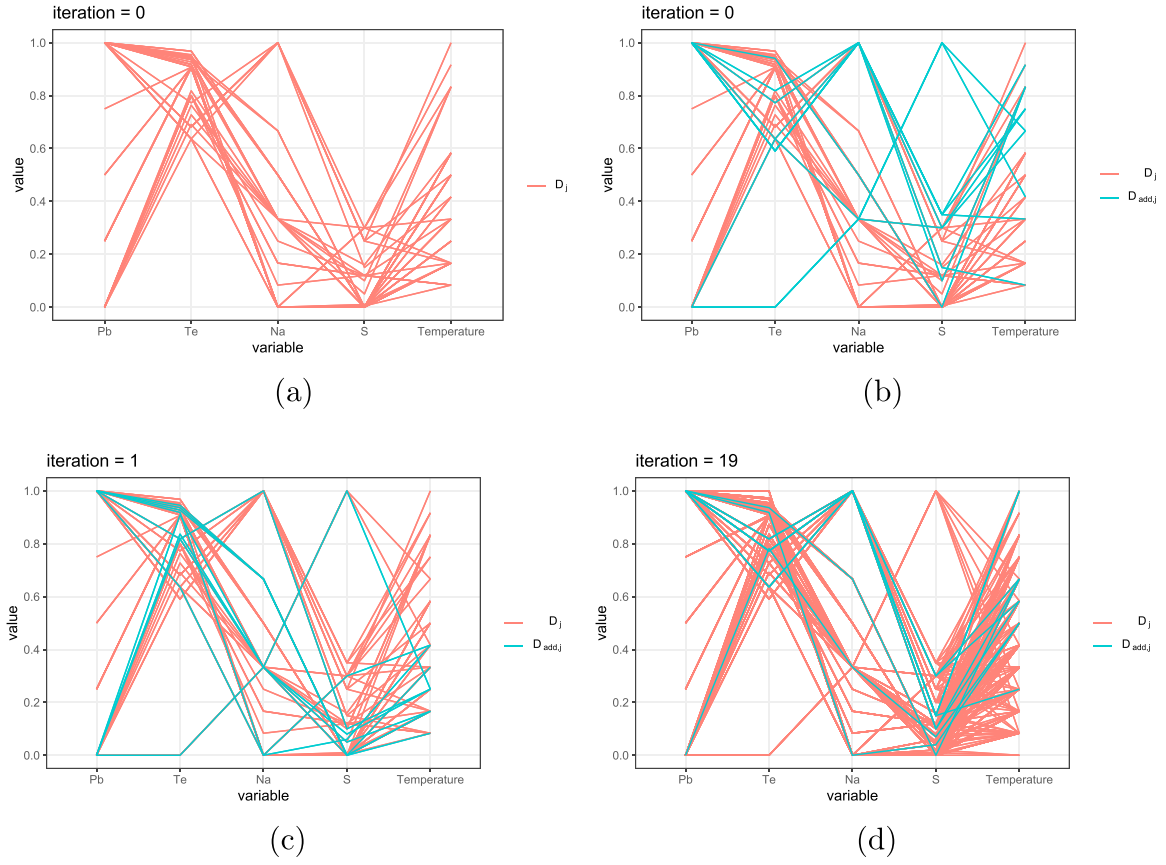


Fig. 10. Parallel coordinates plot over four ALPERC iterations for one of the training-test partitions.

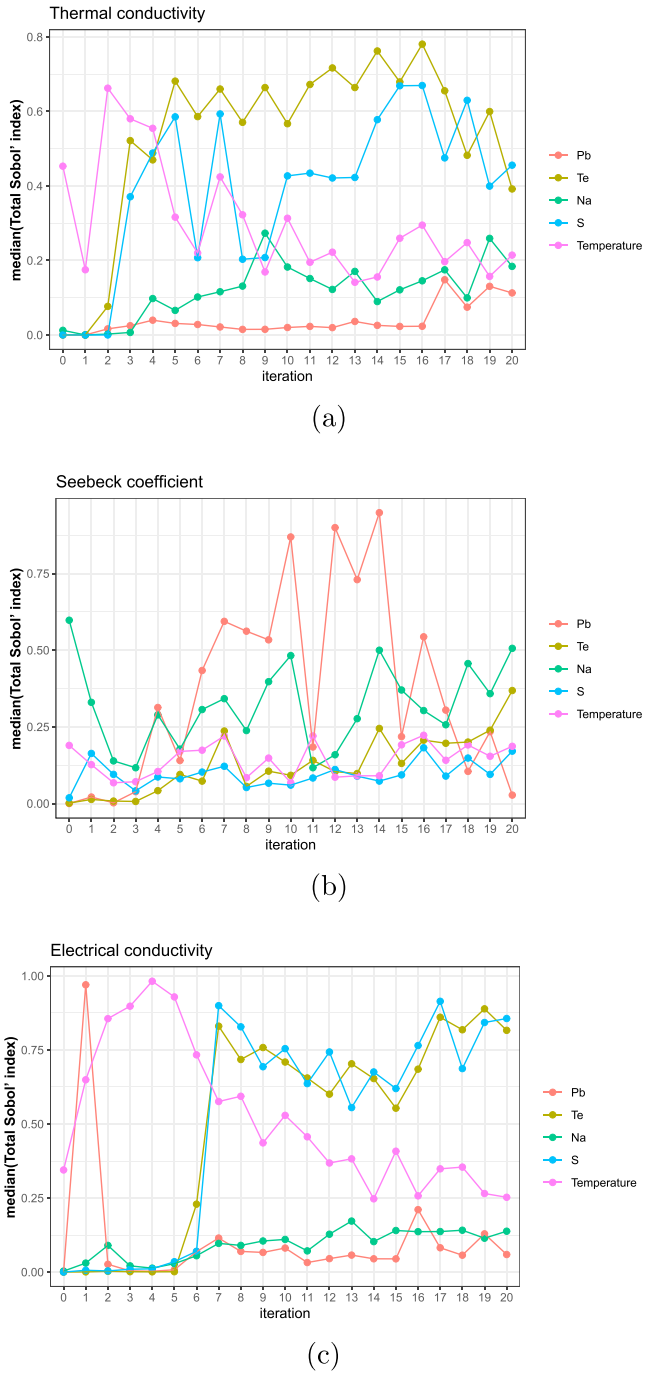


Fig. 11. Median total Sobol' indices over the ALPERC iterations in the case study.

attributing high importance to this predictor. As more data is added sequentially, and more levels of the other factors are explored, the models improve and better estimate the effect of the variables. This stresses the importance of appropriate experimental designs for data collection.

One final point concerns the procedure's computational burden. The calculation of ranks is by far the most computationally intensive operation in ALPERC, and with the adopted settings and 500 permutations for the execution of tests, it takes approximately 2 min for one AL iteration to complete on a mid-range laptop and a sequential implementation. While this is not required by the other AL procedures, we believe that this time is still insignificant when considering the bigger picture that includes the execution of experimental trials.

8. Conclusions

In this paper we have introduced ALPERC, a novel AL algorithm suitable for application to physical experimentation when more than two responses are investigated in the same experiment. Although in this paper we have only used ALPERC with GP models, the algorithm could in principle be used in combination with several ML models given that they provide a quantification of uncertainty. We devote this extension to future developments of the method. The procedure works best if a quantification of variable importance can be computed. Therefore, ALPERC can also be regarded as an interpretable methodology, as it drives the selection of experimental configurations not only by a measure of uncertainty, but also by a quantification of variable importance. This feature can help build trust in the system and, together with the nature of DOE data that consists in randomized controlled experiments, provide a basis for causal claims driven by the algorithmic results.

Using simulations that considered different noise settings and sparsity levels, we proved that ALPERC provides lower average prediction error than other competitors from the literature. This appears most clearly in challenging situations when not all the factors in the experiment are active and the noise is not constant. Another result was that it is always preferable to proceed sequentially rather than to execute experiments in one go. The performance of ALPERC was also tested in a case study on the prediction of the thermal conductivity, Seebeck coefficient and electrical conductivity of thermoelectric materials, highlighting the potential of this method in terms of reduction in global predictive error and interpretability using variable importance.

ALPERC can be used to foster data-driven innovation in several industries where statistically designed experiments are the principal tool for data collection, ensuring accurate and reliable results while limiting the efforts in terms of resources, time and cost.

Author statement

Rosa Arboretti: Validation, Visualization, Resources, Writing - review & editing, Supervision, Project administration. **Riccardo Ceccato**: Validation, Visualization, Software, Writing - review & editing. **Luca Pegoraro**: Conceptualization, Methodology, Validation, Visualization, Investigation, Software, Writing - original draft, Writing - review & editing. **Luigi Salmaso**: Conceptualization, Validation, Resources, Writing - review & editing, Supervision, Project administration.

Data availability statement

The thermoelectric data can be downloaded free of charge from www.starrydata2.org.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

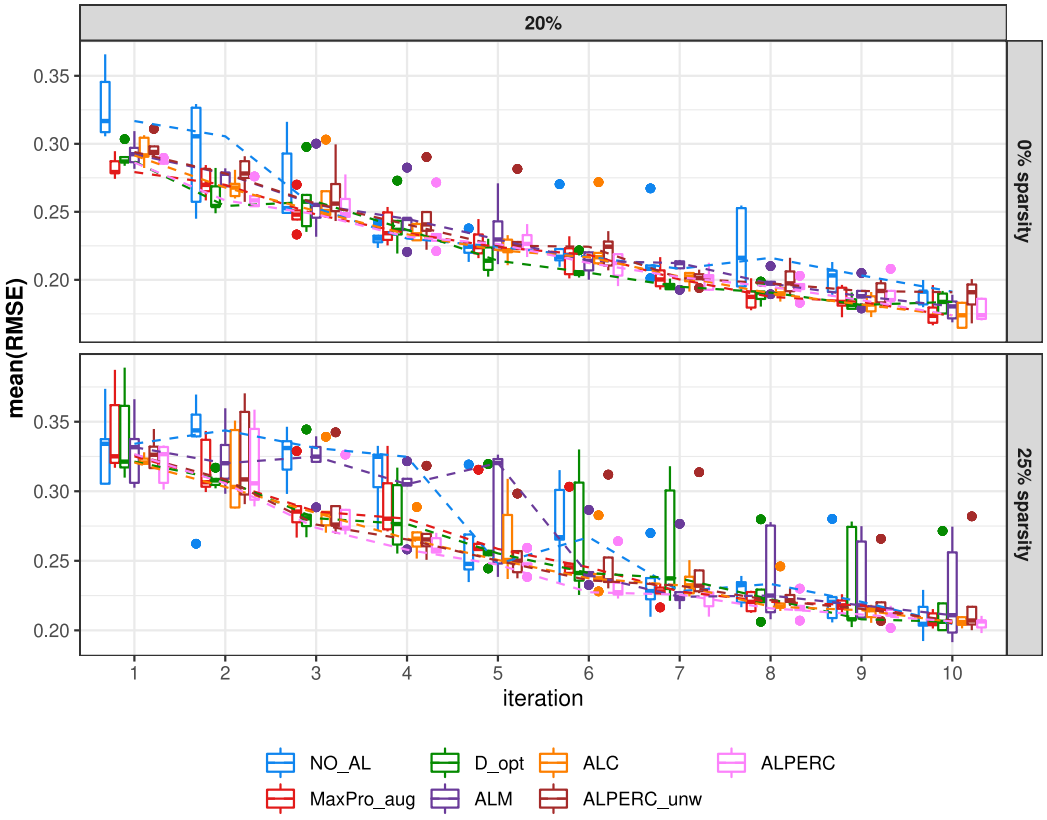


Figure A.12
Simulation results of the average RMSE at each AL iteration for the 20% homoscedastic noise level. The results at *iteration* = 0 are not displayed as they are the same for all the AL strategies.

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